

Chacha Chen

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I am interested in the broad area of human-centered machine learning, at the intersection of machine learning and human-computer interaction. I am passionate to work on topics but not limited to: interpretable ML, human-AI decision making, and human-AI cooperation.

EDUCATION **University of Chicago** 2021 – present
 Department of Computer Science
 Ph.D. in Computer Science
 Advisor: [Chenhao Tan](#)

Pennsylvania State University 2019 – 2020
 College of Information Sciences and Technologies
 Master in Informatics
 Advisor: [Ting-Hao \(Kenneth\) Huang](#), [Zhenhui \(Jessie\) Li](#)

Shanghai Jiao Tong University 2015 – 2019
 Department of Computer Science
 Bachelor of Science in Computer Science

PROFESSIONAL **Amazon AWS**, Applied Scientist Intern Jun. 2022 – Aug. 2022
EXPERIENCE
 IQVIA, Machine Learning Research Intern Jun. 2020 – Aug. 2020

PUBLICATIONS  [Google scholar](#)

Manuscripts and Pre-prints

- P.1 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan.
Machine Explanations and Human Understanding
under review.
- P.2 Vivian Lai, **Chacha Chen**, Alison Smith-Renner, Vera Liao, and Chenhao Tan.
Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies
In arXiv:2112.1147.

Conference Papers

- P.3 Chi-Yang Hsu, Yun-Wei Chu, Vincent Chen, Kuan-Chieh Lo, **Chacha Chen**, Ting-Hao Huang, Lun-Wei Ku. Learning to Rank Visual Stories From Human Ranking Data. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (**ACL 2022**).
- P.4 Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Tao Wen, Zhenhui Li. Knowledge-based Residual Learning. In Proceedings of the The 30th International Joint Conference on Artificial Intelligence (**IJCAI 2021**).
- P.5 **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao, UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data, In Proceedings of The Web Conference 2021 (**WWW 2021**).

- P.6 Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**).
- P.7 Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**).
Best Applied Data Science Paper Award
- P.8 **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**).
- P.9 Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**).
- P.10 Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**).

Workshop Papers

- W.1 Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter, What Makes A Good Reference Manager? A Quantitative Analysis of Bibliography Management Applications, Asian CHI Symposium 2021 (**AsianCHI2021**).
Best Paper Award
- W.2 Jiaqi Wang, Hua Shen, **Chacha Chen**, Frank E. Ritter, Are Learners Satisfied with their MOOC Experiences? Assessing and Improving Online Learners' Interactions, Asian CHI Symposium 2021 (**AsianCHI2021**).
- W.3 **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).
- W.4 **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In Proceedings of the Workshop on Noisy User-generated Text (**W-NUT, EMNLP 2020**).

COMMUNITY ACTIVITIES & SERVICE	Conference Program Committee Member/Reviewer	
	• AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21) • W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020 • NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning	
TEACHING EXPERIENCE	Pennsylvania State University	
	Teaching Assistant IST 261 (Java Course): Application Development Design Studio I Prof. Kenneth Huang	Fall 2020
AWARDS	AAAI-20 VISA Research Scholarship	Feb. 2020
	AAAI-20 Student Travel Grant	Feb. 2020
	Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i>	Aug. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016